# FINAL PROJECT REPORT

## Automated Incident Management Lifecycle

Using ServiceNow Incident Management

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**Platform:** ServiceNow

**Project Version:** 1.0

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*Incident Management | Service Operations Workspace*

# TABLE OF CONTENTS

**1. INTRODUCTION**

**1.1 Project Overview**

**1.2 Purpose**

**2. IDEATION PHASE**

**2.1 Problem Statement**

**2.2 Empathy Map Canvas**

**2.3 Brainstorming**

**3. REQUIREMENT ANALYSIS**

**3.1 Customer Journey Map**

**3.2 Solution Requirement**

**3.3 Data Flow Diagram**

**3.4 Technology Stack**

**4. PROJECT DESIGN**

**4.1 Problem Solution Fit**

**4.2 Proposed Solution**

**4.3 Solution Architecture**

**5. PROJECT PLANNING & SCHEDULING**

**5.1 Project Planning**

**6. FUNCTIONAL AND PERFORMANCE TESTING**

**6.1 Performance Testing**

**7. RESULTS**

**7.1 Output Screenshots**

**8. ADVANTAGES & DISADVANTAGES**

**9. CONCLUSION**

**10. FUTURE SCOPE**

**11. APPENDIX**

# 1. INTRODUCTION

## 1.1 Project Overview

The project, “Automated Incident Management Lifecycle using ServiceNow,” is a ServiceNow-based solution designed to standardize and automate the complete incident lifecycle. It enables Service Desk agents to create and classify incidents, use Agent Assist knowledge, reassign incidents to Level 2 support, initiate emergency change requests, create child incidents, document cause and resolution, and generate knowledge aincidents.

### Purpose:

The purpose is to provide a controlled, transparent, and traceable incident process inside Service Operations Workspace. The solution connects incident intake, classification, assignment, investigation, change integration, child-incident tracking, resolution, knowledge creation, and SLA visibility.

- Enable Service Desk agents to create and classify incidents consistently.
- Integrate Agent Assist and Knowledge Management for faster resolution.
- Reassign incidents to appropriate Level 2 support teams with clear ownership.
- Create emergency change requests directly from incidents when required.
- Create child incidents for recurring or related issues.
- Document probable cause, resolution code, and resolution notes.
- Generate Knowledge Articles from resolved incidents.
- Improve SLA compliance and service visibility.

## 2.2 Empathy Map Canvas

Service Desk agent perspective across incident intake, classification, investigation, resolution and knowledge reuse.

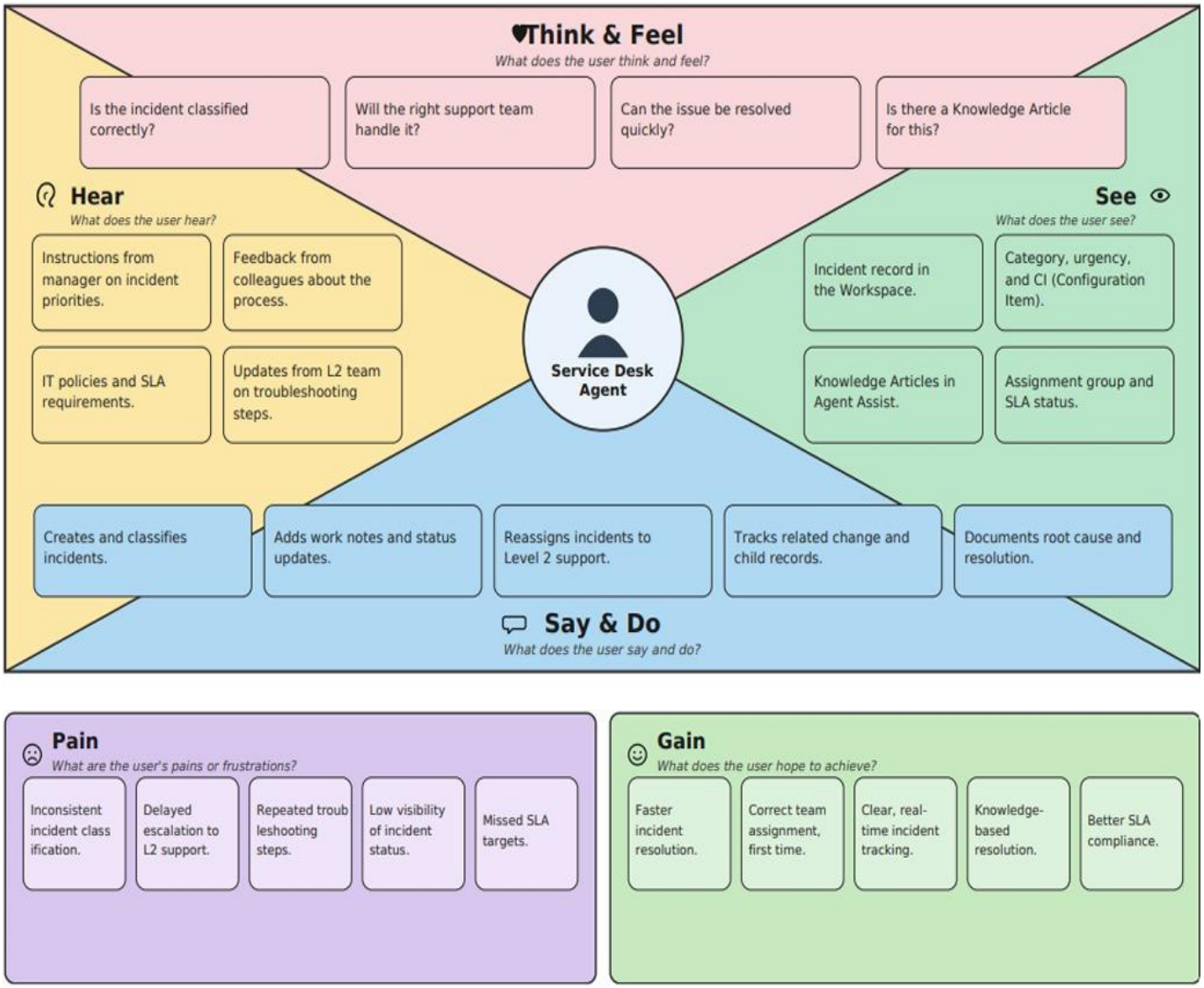



***Figure 1. Empathy Map – Service Desk Agent / Incident Management***

The agent seeks fast classification, clear ownership, useful knowledge, controlled changes, traceable resolution and reliable SLA visibility.

# 2.3 Customer Journey Map

The journey follows a Service Desk incident from recognition through classification, assistance, investigation, resolution and knowledge reuse.

| Journey Stages | 1. Awareness<br>Recognize the incident | 2. Classification<br>Classify the incident | 3. Knowledge & Assignment<br>Find help & assign | 4. Investigation & Change<br>Diagnose & act | 5. Resolution<br>Resolve & document |
|---|---|---|---|---|---|
| Steps<br>What does the user do at this stage? | • Identify the incident and confirm user impact. | • Open the incident in Service Operations Workspace and capture details. | • Review Agent Assist knowledge and route to the right support group. | • Investigate the issue, update the CI, and create an emergency change when required. | • Document cause and resolution, resolve the incident, and create knowledge. |
| Interactions<br>Who or what do they interact with? | • End user<br>Incident record<br>Service Desk | • ServiceNow Workspace<br>Category, urgency, service and CI fields | • Agent Assist<br>Knowledge articles<br>Network / Level 2 team | • Level 2 technicians<br>Change Management<br>Related Records | • Incident record<br>Knowledge Management<br>SLA / Activity history |
| Goals & Motivations<br>What is the user trying to achieve? | • Understand the issue and record it correctly. | • Classify the incident accurately and avoid rework. | • Find a useful solution and ensure clear ownership. | • Restore service quickly with controlled change and documentation. | • Close the incident correctly and preserve reusable knowledge. |
| Positive Moments<br>What do they find enjoyable or satisfying? | • Clear incident intake and complete information. | • Consistent classification and visible assignment. | • Relevant knowledge articles and quick escalation. | • Clear technical ownership and traceable change. | • Successful resolution, knowledge creation, and SLA visibility. |
| Negative Moments<br>What do they find frustrating or confusing? | • Incomplete information or unclear issue description. | • Incorrect category, CI, or urgency causing rework. | • No relevant article or delayed reassignment. | • Repeated troubleshooting, delayed change, or unclear ownership. | • Incomplete resolution notes or poor visibility of related records. |
| Areas of Opportunity<br>How can we make this better? | • Standardize incident intake and required information. | • Use guided classification for category, urgency, service and CI. | • Improve Agent Assist relevance and make escalation structured. | • Link emergency changes and child incidents directly to the incident. | • Generate knowledge from resolutions and strengthen SLA tracking. |
| Emotional Journey<br>How do they feel throughout the journey? | Concerned | Hopeful | Waiting | Frustrated | Satisfied |

**Journey Insight**

- Positive progression comes from accurate classification, relevant Agent Assist knowledge, clear ownership, and traceable resolution.
- Friction occurs when information is incomplete, escalation is delayed, troubleshooting is repeated, or related records are not linked.
- The opportunity is to standardize the lifecycle while keeping change, child incidents, knowledge and SLA records connected.

## 3.2 Solution Requirement

### Functional Requirement

| FR No. | Functional Requirement | Sub Requirement / Expected Behavior |
|---|---|---|
| FR-1 | Incident Record Creation | Create incident in Service Operations Workspace and capture caller and short description. |
| FR-2 | Incident Classification | Capture category, subcategory, urgency, service, service offering, and CI. |
| FR-3 | Agent Assist Knowledge | Show relevant Knowledge Articles based on incident details; attach and mark useful articles. |
| FR-4 | Reassignment & Escalation | Reassign incident to Level 2 support and maintain clear task ownership. |
| FR-5 | Emergency Change | Create and link an emergency change request directly from the incident when needed. |
| FR-6 | Child Incident | Create related child incidents for recurring or related issues and track them. |
| FR-7 | Resolution | Document probable cause, resolution code, and resolution notes; resolve incident. |
| FR-8 | Knowledge Creation | Create a Knowledge Article from a resolved incident using the selected template. |
| FR-9 | SLA & Related Records | Track SLA status and verify linked changes, children, and knowledge records. |

### Non-functional Requirements

| NFR No. | Requirement | Description |
|---|---|---|
| NFR-1 | Usability | Workspace and incident forms should be clear and easy for agents to use. |
| NFR-2 | Security | Only authorized users and groups should update incidents and related records. |
| NFR-3 | Reliability | Incident, change, child, knowledge, and SLA relationships should remain consistent. |
| NFR-4 | Performance | Knowledge, assignment, and related-record operations should respond within acceptable time. |
| NFR-5 | Traceability | Activity history should preserve work notes, assignment, resolution, and linked records. |
| NFR-6 | Scalability | The process should support additional support groups, services, and knowledge content. |

## 3.3 Data Flow Diagram

The data flow shows how an IT incident moves through ServiceNow from intake to classification, knowledge assistance, escalation, change/child tracking, resolution and SLA visibility.




### Data Flow Details

| Source / Actor | Process | Data Flow | Record / Output |
|---|---|---|---|
| End User | Incident Reporting | Issue details, caller information | incident [INC] |
| Service Desk | Classification | Category, urgency, service, CI | incident [INC] |
| Agent Assist | Knowledge Support | Incident context / short description | Knowledge Article |
| Level 2 Support | Investigation | Work notes, CI updates | incident [INC] |
| Change Management | Emergency Change | Change details from incident | change_request [CHG] |
| Incident / Support | Child Incident | Related or recurring issue | incident [Child] |
| Resolution | Knowledge / SLA | Cause, resolution, SLA data | incident / Knowledge / SLA |

### Data Flow Summary

End User → Service Desk → Incident Classification → Agent Assist → Level 2 Investigation → Emergency Change / Child Incident → Resolution & Knowledge Creation → SLA Tracking.

## 3.4 Technology Stack

| Component | Technology / ServiceNow Feature | Purpose |
|---|---|---|
| User Interface | Service Operations Workspace | Agent-facing incident creation, classification, and tracking. |
| Incident Management | Incident module / incident table | Stores and manages incident lifecycle. |
| Knowledge | Agent Assist / Knowledge Management | Finds, attaches, and creates reusable knowledge. |
| Assignment | Assignment Groups / Assigned To | Supports Service Desk and Level 2 ownership. |
| Change Integration | Emergency Change Request | Links change activity to incident resolution. |
| Related Records | Child Incidents / Related Lists | Tracks related and recurring incidents. |
| SLA | Task SLA / SLA related records | Provides service-level tracking and visibility. |
| Configuration | Service / Service Offering / CI | Connects incident to business service and technical asset. |
| Administration | ServiceNow platform configuration | Controls forms, workflows, access, and related records. |

# 4. PROJECT DESIGN

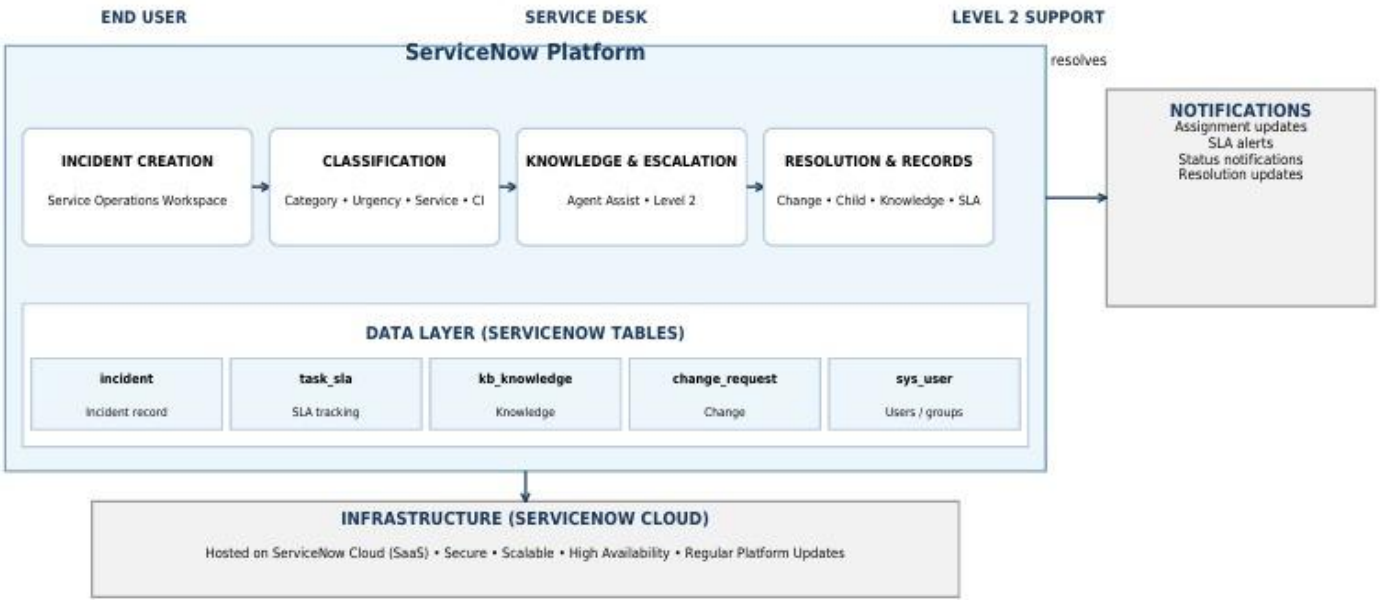
### 4.1 Problem Solution Fit

The problem-solution fit is achieved by replacing fragmented incident coordination with a standardized ServiceNow lifecycle. Agents receive structured classification fields, relevant knowledge, clear escalation, linked changes and child incidents, documented resolution, and reusable knowledge.

| Canvas Area | Project-Specific Definition |
|---|---|
| Customer Segment | Service Desk agents and Level 2 support teams handling IT incidents. |
| Job-to-be-Done | Classify, investigate, escalate, resolve, and document incidents efficiently. |
| Root Cause | Manual coordination, incomplete classification, disconnected knowledge and change activities. |
| Available Solution | Incident module, Agent Assist, assignment groups, change integration, child incidents, Knowledge, SLAs. |
| Constraints | Support-team availability, correct configuration, permissions, and accurate incident data. |
| Solution | Standardized incident lifecycle with integrated related records and knowledge reuse. |

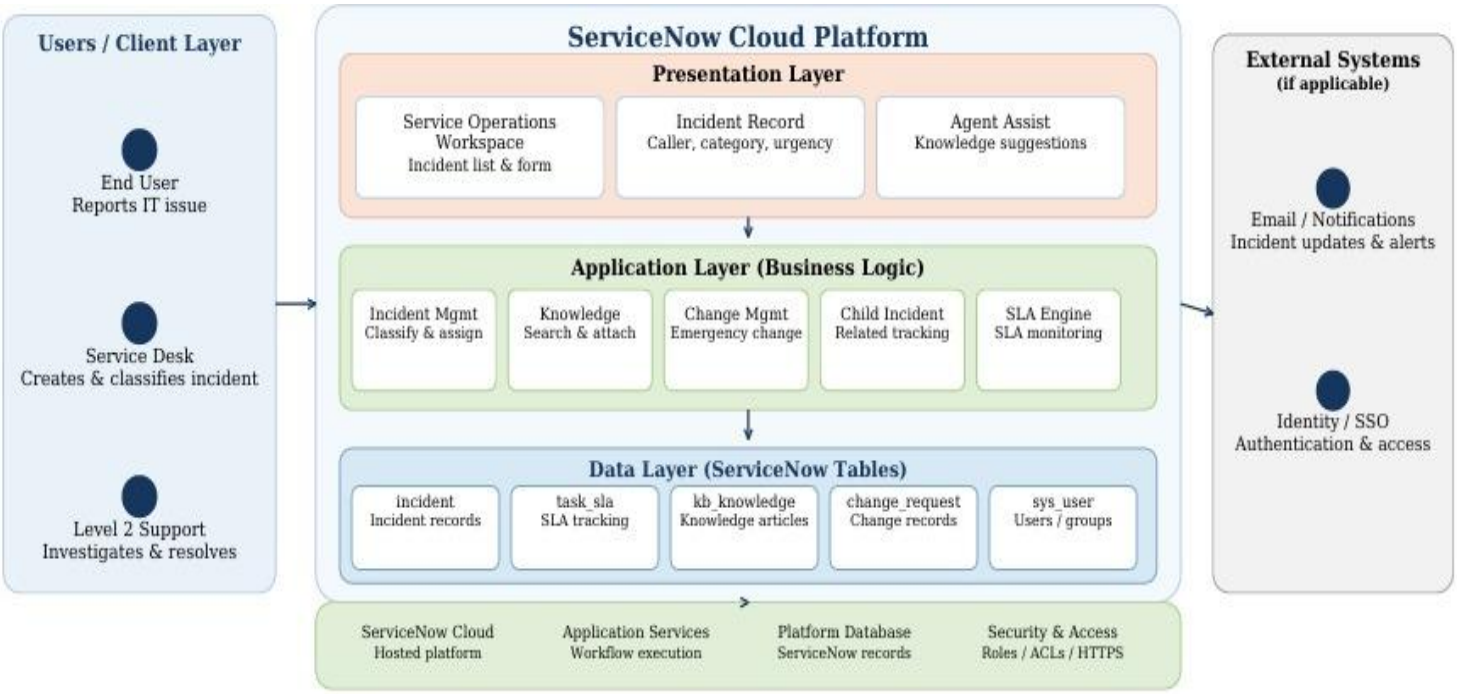
## 4.2 Proposed Solution

1. Create an incident in Service Operations Workspace with caller and short description.
2. Classify the incident using category, subcategory, urgency, service, service offering, and CI.
3. Use Agent Assist to surface relevant Knowledge Articles and attach a useful article.
4. Add watchers and work-note followers where required for communication.
5. Reassign the incident from Service Desk to Network / Level 2 support when advanced investigation is required.
6. Update the Configuration Item and place the incident On Hold with reason Awaiting Change when appropriate.
7. Create an emergency change request from the incident and create a child incident for related/recurring work.
8. Document probable cause, resolution code, and resolution notes, then resolve the parent incident.
9. Create a Knowledge Article from the resolved incident and verify SLA and related-record tracking.



## 4.3 Solution Architecture

The architecture connects the Service Operations Workspace, incident data, Agent Assist, Level 2 support, change/child tracking, resolution, knowledge management and SLA visibility.




### Architecture Components

| Component | Purpose |
|---|---|
| Service Desk / Workspace | Creates and classifies incident records and maintains initial ownership. |
| Incident Data Layer | Stores caller, category, urgency, service, CI, state, notes, and assignment. |
| Agent Assist / Knowledge | Surfaces relevant articles and supports attachment of useful guidance. |
| Assignment & Level 2 | Routes incidents to Network or other technical support groups. |
| Emergency Change | Provides controlled change execution for incidents requiring change activity. |
| Child Incident | Tracks related or recurring incidents without losing parent context. |
| Resolution | Captures probable cause, resolution code, notes, and final state. |
| Knowledge Management | Creates reusable articles from resolved incident knowledge. |
| SLA / Related Records | Provides service-level visibility and verifies linked activities. |

# 5.1 Project Planning

The project is organized into four implementation sprints covering configuration, integration, validation and final lifecycle testing.

| Sprint | User Story | Points | Planned Scope |
|---|---|---|---|
| Sprint-1 | US-01 Incident creation & services | 3 | Create service, service offering, incident intake and required fields. |
| Sprint-1 | US-02 Classification | 3 | Category, subcategory, urgency, service, CI, and work notes. |
| Sprint-2 | US-03 Knowledge & Assignment | 3 | Agent Assist, article attachment, watchers, and Level 2 routing. |
| Sprint-2 | US-04 Change Integration | 3 | Emergency change request and related record verification. |
| Sprint-3 | US-05 Child Incident | 2 | Create and verify child incident tracking. |
| Sprint-3 | US-06 Resolution | 3 | Cause, resolution code, notes, and parent/child resolution. |
| Sprint-4 | US-07 Knowledge Creation | 2 | Create Knowledge Article from resolved incident. |
| Sprint-4 | US-08 SLA & Final Validation | 3 | Validate SLA and complete end-to-end lifecycle. |

## Milestones

1. Incident Record Creation
2. Incident Classification
3. Knowledge Integration
4. Reassignment & Escalation
5. Change Request Creation
6. Child Incident Creation
7. Incident Resolution
8. Knowledge Creation
9. SLA & Related Record Validation

## Skills

- Incident Management
- Agent Assist
- Change Management Integration
- Related Lists
- SLAs
- Knowledge Management

### Lifecycle Coverage

- Create and classify incidents with complete context.
- Use knowledge and structured escalation for faster diagnosis.
- Integrate emergency changes and child incidents.
- Document cause and resolution, then create reusable knowledge.
- Validate SLA status and related records end to end.

# 6. FUNCTIONAL AND PERFORMANCE TESTING

### 6.1 Performance Testing

Validation focuses on consistent processing of the incident lifecycle, related records, knowledge, assignment, and SLA information. The test cases confirm that configured operations work without missing relationships or workflow breaks.

| S.No. | Parameter | Validation | Result |
|---|---|---|---|
| 1 | Incident Creation | Incident is created with caller and short description. | Pass |
| 2 | Classification | Category, subcategory, urgency, service, offering and CI are retained. | Pass |
| 3 | Agent Assist | Knowledge Articles appear and can be attached. | Pass |
| 4 | Assignment | Assignment group changes to Network and Assigned To clears when required. | Pass |
| 5 | Change | Emergency change can be created and linked to the incident. | Pass |
| 6 | Child Incident | Child incident is created and appears in Related Records. | Pass |
| 7 | Resolution | Cause, resolution code, notes and state are saved. | Pass |
| 8 | Knowledge | Knowledge Article is created from resolved incident. | Pass |
| 9 | SLA | SLA and related-record tracking remain visible. | Pass |

### End-to-End Validation

The complete lifecycle was structured for validation as: Incident Creation → Classification → Knowledge Assistance → Reassignment → Emergency Change / Child Incident → Cause & Resolution → Resolve → Knowledge Creation → SLA Verification.

# 7. RESULTS

**Functional validation summary**

| Test Area | Expected Outcome | Status |
|---|---|---|
| Incident Creation | Incident record is accessible and required details can be submitted. | Pass |
| Classification | Incident fields retain the required service and technical context. | Pass |
| Knowledge & Agent Assist | Relevant articles can be opened, marked helpful, and attached. | Pass |
| Level 2 Assignment | Incident ownership moves to the correct support group. | Pass |
| Change Integration | Emergency change is linked to the incident. | Pass |
| Child Incident | Related child incident is visible from the parent. | Pass |
| Resolution | Cause and resolution documentation are retained. | Pass |
| Knowledge Creation | Resolved incident can generate a Knowledge Article. | Pass |
| SLA & Related Records | SLA and linked records remain visible. | Pass |

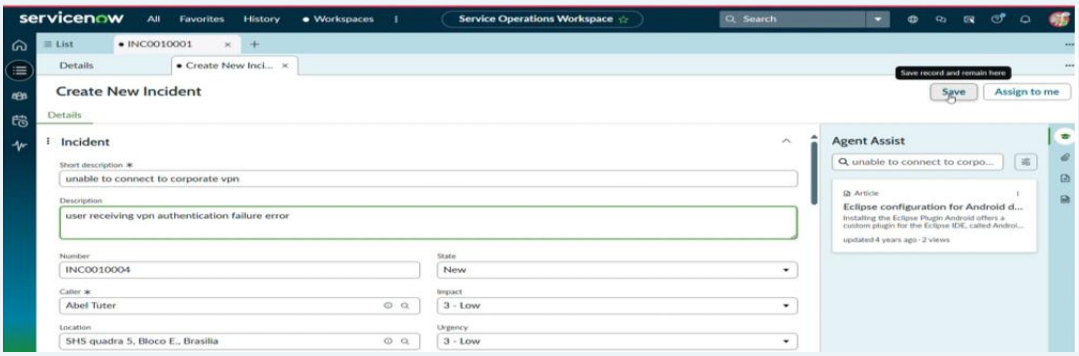
## 7.1 Output Screenshots

The final submission should include ServiceNow evidence for each major lifecycle stage. The following output list is aligned with the implemented project flow.

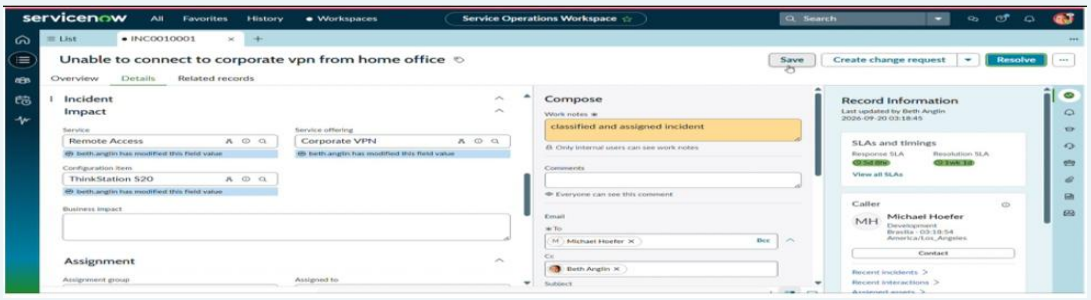
- Output 1 – Incident created in Service Operations Workspace.
- Output 2 – Incident classification and Service / Service Offering / CI.
- Output 3 – Agent Assist knowledge article attached and marked helpful.
- Output 4 – Watch list and work-note followers.
- Output 5 – Reassignment to Network / Level 2 support.
- Output 6 – Configuration Item updated and On Hold / Awaiting Change.
- Output 7 – Emergency Change Request linked to incident.
- Output 8 – Child Incident listed under Related Records.
- Output 9 – Cause, resolution code, resolution notes and Resolved state.
- Output 10 – Knowledge Article created and SLA visible.

# 7.1 Output Screenshots – Incident Lifecycle

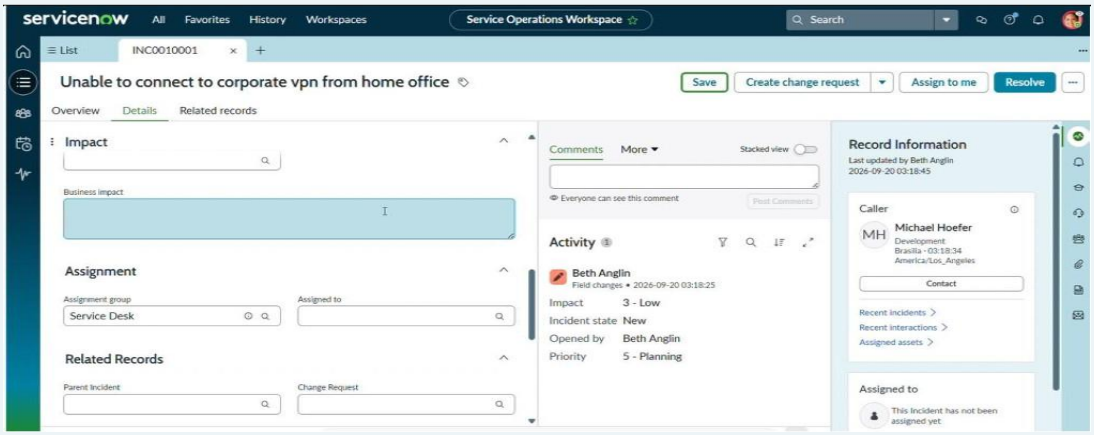
**1. Incident creation / Agent Assist**




**2. Incident classification details**

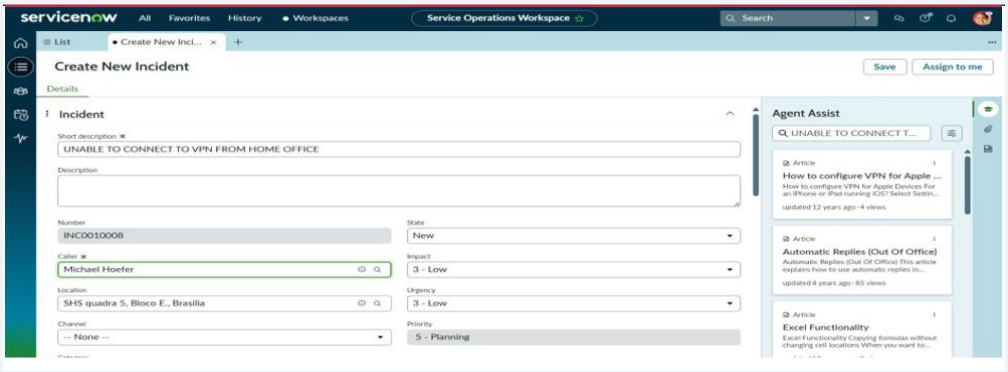



**3. Assignment to Service Desk**

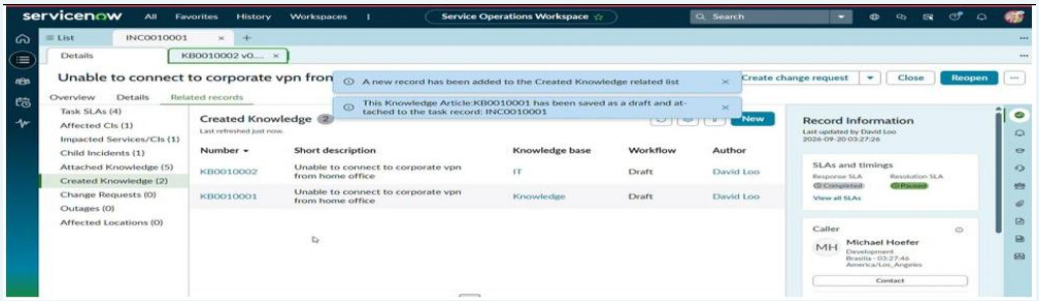


# 7.1 Output Screenshots – Resolution & Knowledge

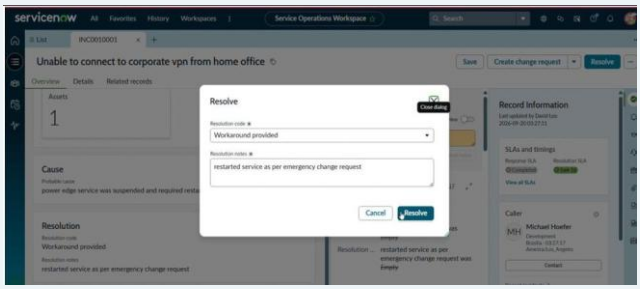
**4. Incident creation with Agent Assist**



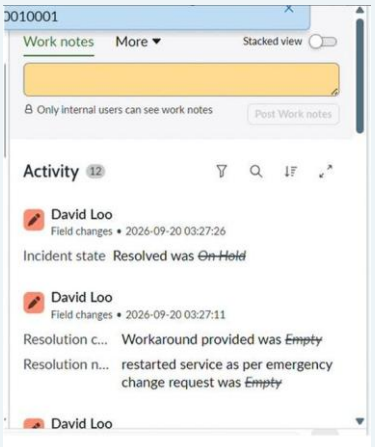

**5. Knowledge article created and linked**




**6. Resolution dialog / workaround**




**7. Activity history showing resolution**



# 8. ADVANTAGES & DISADVANTAGES

| Advantages | Disadvantages / Limitations |
|---|---|
| Standardized incident intake and classification. | Solution depends on correct ServiceNow configuration and permissions. |
| Knowledge support can reduce repeated troubleshooting. | Knowledge quality depends on available and maintained articles. |
| Clear Level 2 ownership and escalation. | Resolution may still depend on support-team availability. |
| Emergency change linkage improves traceability. | Emergency changes still require appropriate change governance. |
| Child incidents provide structured tracking for related issues. | Additional related records increase process administration. |
| Cause and resolution documentation improves history and reuse. | Manual data entry is still required for some fields. |
| Knowledge creation turns resolved work into reusable guidance. | Generated knowledge still requires review and maintenance. |
| SLA tracking improves visibility of service performance. | SLA configuration must match organizational policies. |

## Project Impact

- Faster and more standardized incident handling.
- Improved visibility across Service Desk and Level 2 teams.
- Better linkage between incident, change, child incident, knowledge, and SLA records.
- Reusable resolution knowledge for future incidents.
- Clearer ownership and activity history throughout the lifecycle.

# 9. CONCLUSION

The Automated Incident Management Lifecycle project demonstrates how ServiceNow can be configured to standardize incident handling from creation through resolution. The solution connects incident classification, Agent Assist, Level 2 escalation, emergency change integration, child incidents, resolution documentation, knowledge creation, and SLA tracking.

By maintaining related records within the incident lifecycle, the project provides a clearer operational trail for Service Desk and technical support teams. The process also turns successful resolutions into reusable knowledge, supporting continuous improvement.

# 10. FUTURE SCOPE

- Add automated assignment rules based on category, service, CI, or impact.
- Integrate monitoring/event management to create incidents automatically.
- Add richer SLA dashboards and proactive escalation notifications.
- Use AI-assisted incident summarization and resolution recommendation.
- Expand Knowledge Management with article quality and feedback analytics.
- Integrate CMDB health and dependency information for faster root-cause analysis.
- Add automated change-risk checks before emergency change submission.
- Introduce trend analytics for recurring incidents and problem-management handoff.

# 11. APPENDIX

## A. Core ServiceNow Records / Areas Used

| Record / Area | Meaning | Project Use |
|---|---|---|
| incident | Incident record | Stores caller, classification, assignment, work notes, state and resolution. |
| kb_knowledge | Knowledge Article | Stores reusable troubleshooting and resolution guidance. |
| sys_user | User | Supports caller, watcher, follower and assigned-user references. |
| cmdb_ci | Configuration Item | Identifies the technical CI affected by the incident. |
| cmdb_ci_service | Business / Technical Service | Represents Remote Access service. |
| service_offering | Service Offering | Represents Corporate VPN offering. |
| Change Request | Change record | Tracks emergency change required for incident resolution. |
| Child Incident | Related incident | Tracks related or recurring incident work. |
| Task SLA | SLA record | Provides SLA timing and compliance visibility. |

## B. Acceptance Criteria

- Service Desk agent can create an incident with caller and short description.
- Incident classification captures the required service, offering, urgency, category, subcategory and CI.
- Agent Assist shows relevant knowledge and allows an article to be attached.
- Incident can be reassigned to Network / Level 2 support with clear ownership.
- Emergency change can be created and linked from the incident.
- Child incident can be created and verified under Related Records.
- Probable cause, resolution code and resolution notes are recorded before resolution.
- Resolved incident can generate a Knowledge Article.
- SLA and related records remain visible for lifecycle tracking.
- Parent and child incident activity provides a traceable resolution history.

## C. Project Milestones

Incident Record Creation → Incident Classification → Knowledge Integration → Reassignment & Escalation → Change Request Creation → Child Incident Creation → Incident Resolution → Knowledge Creation → SLA & Related Record Validation.